LECTURE NOTES MATH255

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1. Lecture 8

In this class we talk about factorization of polynomials and the division algorithm.

1.1. **Division algorithm for polynomials.**

Let F be a field and let $F[x]$ be the ring of polynomials in x and coefficients in F. Let $a_0 + ... + a_n x^n = g(x) \in F[x]$ be a polynomial of degree n (i.e., $a_n \neq 0$ and $a_{n+k} = 0$ for all $k \geq 1$). We note that if $m = n + k \geq n$ then we can always write:

$$\begin{aligned} x^{n+k} =& x^k x^n = x^k \cdot \frac{1}{a_n} \cdot \left(g(x) - a_{n-1}x^{n-1} - ... - a_0\right) = \\ & \left(\frac{x^k}{a_n}\right) \cdot g(x) + \left(\left(\frac{-a_{n-1}}{a_n}\right) x^{n+k-1} + ... + \left(\frac{a_0}{a_n}\right)\right) \end{aligned}$$

that is, we can write $x^m = q_1(x)g(x) + r_1(x)$ where $r_1(x)$ has degree $m-1$. By applying the same argument to x^{m-1} (the highest order term of $r_1(x)$), if $m-1 \geq n$, we can write $r_1(x) = q_2(x)g(x) + r_2(x)$ where $r_2(x)$ has degree $m-2$. We can proceed by induction until $m - j < n$, and we obtain an expression:

$$x^m = q(x)g(x) + r(x) \tag{1.1}$$

where the degree of $r(x)$ is less than n. This observation essentially shows the following *division algorithm* for polynomials:

Theorem 1.1. *Let $f(x), g(x) \in F[x]$ then there are unique polynomials $q(x), r(x) \in F[x]$ such that $f(x) = q(x)g(x) + r(x)$ and $\deg(r(x)) < \deg(g(x))$.*

Proof. Let $\deg(g(x)) = n$ and write:

$$f(x) = c_k x^{n+k} + ... + c_0 x^n + b_{n-1}x^{n-1} + ... + b_0. \tag{1.2}$$

By the proceeding calculation we find for each $0 \leq j \leq k$ polynomials $q_j(x), r_j(x) \in F[x]$ such that $\deg(r_j(x)) < n$ and $x^{n+j} = q_j(x)g(x) + r_j(x)$. It follows that we can write:

$$\begin{aligned} f(x) =& c_k \left(q_k(x)g(x) + r_k(x)\right) + ... + c_0 \left(q_0(x)g(x) + r_0(x)\right) + b_{n-1}x^{n-1} + ... + b_0 = \\ & \left(c_k q_k(x) + ... + c_0 q_0(x)\right) g(x) + \left(c_k r_k(x) + ... + c_0 r_0(x) + b_{n-1}x^{n-1} + ... + b_0\right) \end{aligned}$$

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so if we define:

$$q(x) = c_k q_k(x) + ... + c_0 q_0(x), \tag{1.3}$$

$$r(x) = c_k r_k(x) + ... + c_0 r_0(x) + b_{n-1}x^{n-1} + ... + b_0 \tag{1.4}$$

then $f(x) = q(x)g(x) + r(x)$ and since every summand of $r(x)$ has degree $< n$ we conclude that $r(x)$ has degree smaller than n. It remains to show uniqueness of $q(x)$ and $r(x)$. Assume that we could write $f(x) = q(x)g(x) + r(x) = q'(x)g(x) + r'(x)$ with $r(x), r'(x)$ both having degree smaller than n. Then we have:

$$q(x)g(x) + r(x) = q'(x)g(x) + r'(x) \iff (q(x) - q'(x))g(x) = r'(x) - r(x) \tag{1.5}$$

but if $q(x) - q'(x) \neq 0$ then the degree of $(q(x) - q'(x))g(x)$ is at least $\deg(g(x)) = n$ (since F has no zero divisors the product of the highest order terms of two polynomials are non-zero), which would be a contradiction since the right hand side has degree at most $n - 1$. We conclude that $q(x) - q'(x) = 0$ or $q(x) = q'(x)$. It then follows that $r'(x) - r(x) = 0$ as well so $r'(x) = r(x)$. ■

1.2. **Factors.** Let F be a field and $F[x]$ the corresponding ring of polynomials. If $f(x) \in F[x]$ can be written $f(x) = g(x)h(x)$, $g(x), h(x) \in F[x]$, then we say that $g(x)$ and $h(x)$ are factors of $f(x)$.

Theorem 1.2. *Let $f(x) \in F[x]$ and $a \in F$. The polynomial $g(x) = x - a$ is a factor of $f(x)$ if and only if a is a zero for $f(x)$.*

Proof. If $g(x) = x - a$ is a factor of $f(x)$ then we can write $f(x) = g(x)h(x)$ for some $h(x) \in F[x]$. We apply the evaluation homomorphism $\phi_a(f(x)) = \phi_a(g(x)h(x)) = \phi_a(g(x))\phi_a(h(x))$ and note that $\phi_a(g(x)) = a - a = 0$, so $\phi_a(f(x)) = 0 \cdot \phi_a(h(x)) = 0$, that is a is a zero for $f(x)$.

Conversely, assume that a is a zero for $f(x)$. By the division algorithm we can write $f(x) = q(x)g(x) + r(x)$ where $r(x) = b$ is constant (since $g(x)$ has degree 1 we must have that $r(x)$ has degree at most 0, i.e., $r(x)$ must be constant). Recalling that a is a zero of $f(x)$, we apply the evaluation homomorphism ϕ_a on both sides and obtain:

$$0 = \phi_a(f(x)) = \phi_a(q(x)g(x) + b) = \phi_a(q(x))\phi_a(g(x)) + b = b \tag{1.6}$$

where we have used that $\phi_a(g(x)) = 0$, and that $\phi_a(b) = b$ since b is constant. We conclude that $r(x) = b = 0$ so $f(x) = q(x)g(x)$ which shows that $g(x)$ is a factor of $f(x)$. ■

Theorem 1.3. *If $f(x) \in F[x]$ has degree n then $f(x)$ has at most n zeros in F.*

Proof. Let $a_1, ..., a_N$ be zeros of $f(x)$ in F. By the proceeding theorem we find $q(x) \in F[x]$ such that $f(x) = q(x)(x - a_1)...(x - a_N)$ but $\deg(f(x)) = \deg(q(x)(x - a_1)...(x - a_N)) \geq \deg((x - a_1)...(x - a_N)) = N$ so we see that $N \leq \deg(f(x))$, i.e., the number of zeros of $f(x)$ in F is at most $\deg(f(x))$. ■

An important concept when we factor polynomials is that of irreducible polynomials.

Definition 1.1. *A polynomial $f(x) \in F[x]$ is irreducible over F if its only factors are multiples of $f(x)$ and constants $a \in F$. Equivalently, $f(x)$ is irreducible over F if $f(x)$ can* ***not*** *be written as $f(x) = g(x)h(x)$ where both $g(x)$ and $h(x)$ has degree strictly lower than $f(x)$. A nonconstant polynomial that is not irreducible over F is reducible over F.*

Example 1.1. The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ is not irreducible over $\mathbb{Q}$. Indeed, if it was then it would have a factor which would have to be of degree 1 (we usually call a factor of degree 1 a *linear factor*). That is, we must be able to write $x^2 - 2 = (x-a)(x-b)$ for some $a, b \in \mathbb{Q}$. But this implies that $x^2 - 2$ has zeros at the rational numbers a and b. On the other hand we know that $x^2 - 2$ only has zeros at $\pm\sqrt{2}$ which are not rational numbers, so we conclude that $x^2 - 2$ can not be written as a product of two polynomials with strictly smaller degree that 2, so $x^2 - 2$ is irreducible.

The previous example generalizes to a slightly more general situation:

Theorem 1.4. *Let F be a field and $f(x) \in F[x]$ be a polynomial of degree 2 or 3. Then $f(x)$ is reducible if and only if $f(x)$ has a zero in F.*

Proof. Note that if $f(x)$ has a zero a in F then Theorem 1.2 implies that $x - a$ is a factor of $f(x)$, so $f(x) = q(x)(x-a)$. Notice that $f(x)$ has degree 2 or 3, so $q(x)$ must have degree 1 or 2. We conclude that $f(x)$ is reducible since it can be written as a product of polynomials with strictly smaller degree.

Conversely, if $f(x)$ is reducible then one of the factors have to be linear $x - a$ for some a. Indeed, if we write $f(x) = g(x)h(x)$ where $g(x)$ and $h(x)$ have strictly smaller degree than $f(x)$ then either $g(x)$ has degree 1 and $h(x)$ has degree 2 or $g(x)$ has degree 2 and $h(x)$ has degree 1. In either case, one of the factors of $f(x)$ has degree 1. That is, there is some $a \in F$ such that $f(x) = q(x)(x-a)$ and by Theorem 1.2 we conclude that a is a zero for $f(x)$, so $f(x)$ has a zero in F. ■

Example 1.2. The polynomial $f(x) = 2x^2 + x + 1 \in \mathbb{Z}_3[x]$ is irreducible over $\mathbb{Z}_3$. Since $f(x)$ has order 2 it suffices to show that $f(x)$ has no zero in $\mathbb{Z}_3$. We can do this by hand:

$$f(0) = 2 \cdot 0^2 + 0 + 1 = 1, \tag{1.7}$$

$$f(1) = 2 \cdot 1^2 + 1 + 1 = 4 = 1, \tag{1.8}$$

$$f(2) = 2 \cdot 2^2 + 2 + 1 = 11 = 2. \tag{1.9}$$

Since $f(x)$ has no zero in $\mathbb{Z}_3$, and since $f(x)$ has degree 2 we conclude that $f(x)$ is irreducible over $\mathbb{Z}_3$.

The following theorem is useful sometimes when dealing with integer polynomials, but we will omit its (rather technical) proof.

Theorem 1.5. *Let $f(x) \in \mathbb{Z}[x]$, then $f(x)$ factors into a product of lower degree polynomials in $g(x), h(x) \in \mathbb{Z}[x]$ if and only if $f(x)$ factors into a product of lower degree polynomials in $\tilde{g}(x), \tilde{h}(x) \in \mathbb{Q}[x]$. Moreover, the degrees of the polynomials can be taken to be the same:* $\deg(g(x)) = \deg(\tilde{g}(x))$, *and* $\deg(h(x)) = \deg(\tilde{h}(x))$.

Remark 1. A useful consequence of this theorem is that if we want to show that $f(x) \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$, then it suffices to show that $f(x)$ does not have factors $g(x), h(x) \in \mathbb{Z}[x]$ which is often easier than showing that $f(x)$ does not have factors in $\mathbb{Q}[x]$.

An immediate consequence is that if a polynomial $f(x) \in \mathbb{Z}[x]$ has a root in $\mathbb{Q}$ then it also has a root in $\mathbb{Z}$. Indeed, if $f(x)$ has a root in $\mathbb{Q}$ then $f(x)$ has a linear factor $x - a$, $a \in \mathbb{Q}$, but by the above theorem this implies that $f(x)$ also has a linear factor over $\mathbb{Z}$ which implies that $f(x)$ has a root in $\mathbb{Z}$.

Theorem 1.6 (Eisentstein's criterion)**.** *Let p be a prime. If $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_0 \in \mathbb{Z}[x]$ with a_n not divisible by p, every $a_{n-1}, ..., a_0$ divisible by p, and a_0 not divisible by p^2, then $f(x)$ is irreducible over $\mathbb{Q}$.*

Proof. We will use the map $\hat{\sigma}_p$ from the homework, for a proof that does not use this you can read the book. It suffices to show that there does not exist $g(x), h(x) \in \mathbb{Z}[x]$ such that $f(x) = g(x)h(x)$ and $\deg(h(x)), \deg(g(x)) < \deg(f(x)) = n$. Note that we also have $\deg(g(x)) + \deg(h(x)) = \deg(f(x))$. Assume for contradiction that such $g(x)$ and $h(x)$ exist. Let $\hat{\sigma}_p(f(x)) = \tilde{f}(x)$, $\hat{\sigma}_p(g(x)) = \tilde{g}(x)$, $\hat{\sigma}_p(h(x)) = \tilde{h}(x)$. Then $\tilde{f}(x) = \tilde{g}(x)\tilde{h}(x)$. From the assumptions on the coefficients of $f(x)$ we see that $\tilde{f}(x) = a_n x^n$ with $a_n \neq 0$. We claim that both $\tilde{g}(x)$ and $\tilde{h}(x)$ has zeros at $x = 0$. Indeed, we have: $0 = \tilde{f}(0) = \tilde{g}(0)\tilde{h}(0)$ so, since $\mathbb{Z}_p$ is a field and therefore an integral domain, we conclude that either $\tilde{g}(0) = 0$ or $\tilde{h}(0) = 0$. Assume that $\tilde{g}(0) = 0$, then we can factor $\tilde{g}(x) = x\tilde{q}(x)$ and we obtain an equality, after cancelling x on both sides (recall that $\mathbb{Z}_p[x]$ is an integral domain), an equality $a_n x^{n-1} = \tilde{q}(x)\tilde{h}(x)$. We can now use the same argument again to deduce that either $\tilde{q}(0) = 0$ or $\tilde{h}(0) = 0$, if $\tilde{h}(0) = 0$ then we have shown that both $\tilde{g}(x)$ and $\tilde{h}(x)$ has zeros at $x = 0$. If $\tilde{q}(0) = 0$ then we proceed with the same argument again. After using this argument $\deg(g(x))$ times we will have either showed that $\tilde{h}(0) = 0$ or we arrive at an equality $a_n x^{n-\deg(g(x))} = \tilde{h}(x)$ and since $n = \deg(f(x)) > \deg(g(x))$ this implies that $\tilde{h}(0) = 0$. Let $g(x) = b_s x^s + ... + b_0$ and $h(x) = c_t x^t + ... + c_0$, then we have in $\mathbb{Z}_p$:

$$0 = \tilde{g}(0) = b_0 \mod p,$$
$$0 = \tilde{h}(0) = c_0 \mod p,$$

from which we deduce that b_0 and c_0 are both divisible by p. Finally, note that $a_0 = b_0 c_0$ so, since b_0, c_0 are both divisible by p, we conclude that a_0 is divisible by p^2. This contradicts the assumption that a_0 is not divisible by p^2, and it follows that $f(x)$ can not be written as the product $f(x) = g(x)h(x)$, so $f(x)$ is irreducible. ■

1.3. **Unique factorization.**

We want to end these notes by showing that any polynomial over a field F can be decomposed (essentially uniquely) into a product of irreducible polynomials.

Definition 1.2. *A polynomial $p(x) \in F[x]$ divides a polynomial $q(x) \in F[x]$ if we can write $q(x) = p(x)r(x)$ for some polynomial $r(x) \in F[x]$.*

We will prove the following theorem later in the course, for now we will take it for granted.

Theorem 1.7. *Let $f(x) \in F[x]$ be irreducible over F, and $r(x), s(x) \in F[x]$ polynomials such that $f(x)$ divides $r(x)s(x)$. Then $f(x)$ divides $r(x)$ or $f(x)$ divides $s(x)$.*

Theorem 1.8. *Let $f(x) \in F[x]$ be irreducible over F, and let $g_1(x), ..., g_n(x) \in F[x]$. If $f(x)$ divides the product $g_1(x)g_2(x)...g_n(x)$ then $f(x)$ divides at least one of the factors $g_j(x)$.*

Proof. Apply the previous theorem to the product $[g_1(x)...g_{n-1}(x)]g_n(x)$, if $f(x)$ divides $g_n(x)$ then we are done. If $f(x)$ divides $g_1(x)...g_{n-1}(x)$ then we apply the previous theorem to the product $[g_1(x)...g_{n-2}(x)]g_{n-1}(x)$. Proceeding by induction the theorem follows. ∎

Theorem 1.9. *Let $f(x) \in F[x]$ be any non-constant polynomial. There are irreducible polynomials $p_1(x), ..., p_n(x) \in F[x]$ such that $f(x) = p_1(x)...p_n(x)$. Moreover, the irreducible polynomials $p_1(x), ..., p_n(x)$ are unique up to changing the order of the polynomials, and multiplying them with a unit (i.e., a non-zero element in F).*

Proof. We begin by showing that a non-constant polynomial $f(x) \in F[x]$ can be written as a product of irreducible polynomials. A quick proof of a decomposition into irreducible polynomials is to use induction. Note that any polynomial of degree 1 is irreducible since it can not be written as a product of polynomials of lower degree (any polynomial of lower degree than 1 would be constant). Let $\deg(f(x)) = n$, and assume for induction that we know that any polynomial of degree $\leq n-1$ can be written as a product of irreducible polynomials (using that degree 1 polynomials are irreducible as a base case). If $f(x)$ is irreducible, there is nothing to prove so assume that $f(x)$ is reducible. That is, we assume that we can write $f(x) = g(x)h(x)$ where $g(x)$ and $h(x)$ both have degree lower than $f(x)$. By the induction hypothesis we conclude that $g(x) = p_1(x)...p_m(x)$ with $p_j(x)$ all irreducible, and we can write $h(x) = q_1(x)...q_l(x)$ with all $q_j(x)$ irreducible. Finally, we have:

$$f(x) = g(x)h(x) = p_1(x)...p_m(x)q_1(x)...q_l(x) \tag{1.10}$$

where each $p_1(x), ..., p_m(x), q_1(x), ..., q_l(x)$ are all irreducible. The existence of a decomposition into irreducible polynomials now follows by induction.

It remains to prove the uniqueness claim. Assume that we can write $f(x) = p_1(x)...p_m(x) = q_1(x)...q_l(x)$ with all $p_1(x), ..., p_m(x), q_1(x), ..., q_l(x) \in F[x]$ irreducible. Since $p_1(x)$ divides $f(x) = q_1(x)...q_l(x)$ we can use Theorem 1.8 to deduce that $p_1(x)$ divides some $q_j(x)$. After reordering the polynomials $q_1(x), ..., q_l(x)$ we will assume that $p_1(x)$ divides $q_1(x)$. Since $q_1(x)$ is irreducible it follows that $q_1(x) = up_1(x)$ with $u \in F \setminus 0$. If we substitute this into the equation $p_1(x)...p_m(x) = q_1(x)...q_l(x)$ then we obtain:

$$p_1(x)...p_m(x) = up_1(x)q_2(x)...q_l(x)$$

and since $F[x]$ is an integral domain (from the homework) we can cancel the $p_1(x)$ on both sides and obtain:

$$p_2(x)...p_m(x) = uq_2(x)...q_l(x). \tag{1.11}$$

Assume for simplicity that $m \geq l$ (the other case is completely analogous). Proceeding by induction we obtain $p_j(x) = u_j q_j(x)$ for all $j = 1, ..., m$ where $u_j \in F \setminus 0$. This gives us an equality:

$$1 = u_1 ... u_m \cdot q_{m+1}(x) ... q_l(x) \tag{1.12}$$

from which it follows that $m = l$, and $u_1 ... u_m = 1$. This finishes the proof. ■

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